GAS PROCESSING & METHANOL COMPLEX PROJECT NO. BFPCL-DSV-RAEC 2026

Project	FEED for Gas Gathering Pipelines & Associated Infrastructure
Client	Brass Fertilizer and Petrochemical Company Limited
Originating Company	Epic Atlantic Limited
Document Title	Pipeline Decommissioning & Abandonment Philosophy: OKNU-BBOU P/L
Document Number	EA-BFPCL-DSV-GGPL-FEED-GEN-PHY-007
Document Revision	R01
Document Status	Issued as a Requirement
Endorsement	Aloy Onyenucheya Duru, COREN (R.7580)
Security Classification	Restricted
Issue Date	11 May 2026

Pipeline Decommissioning & Abandonment Philosophy
OKNU-BBOU Pipeline Segment

R01	09 May 2026	Issued for Client Review	Aloy O Duru	Obima Oragwu	Aloy O Duru COREN (R.7580)	
Rev #	Issue Date	Status Description	Originator	Reviewed	Endorsement (Epic Atlantic)	Approval (BFPCL)

Revision History

Rev. No.	Date	Description of Revision
01	11 May 2026	To accord OKNU-BBOU a status of its own in the D & A exercise, distinct from the integrated OKNU-BBOU-ELEP-AKAB pipeline, as it operates under an independent Gas Sale and Purchase Agreement (GSPA) and project status

Revision Philosophy:

a. All documents for review shall be issued at R01, with subsequent R02, R03, etc as required.
b. All documents approved for issue or approved for design shall be issued at A01 with subsequent A02, A03, etc. as required. (Management of Change is required for A02, A03, etc.)
for the subsequent project phases, post FEED, it is expected that:
c. All Detailed Design documents for review shall be issued at D01, with subsequent D02, D03, etc. as required.
d. All documents approved for construction shall be issued at C01 with subsequent C02, C03, etc. as required. (Management of Change is required for C02, C03, etc.)
e. All approved "As-Built" documents shall be issued at Z01, with subsequent Z02, Z03, etc as required. (Use versions Z01.1, Z01.2, Z01.3, etc to review the "As-Built" document to Z02).

Summary Page

Client:	Brass Fertilizer & Petrochemical Company Limited	
Project:	Gas Processing & Methanol Complex	**Project No.** BFPCL-DSV-RAEC 2026
Facility:	Gas Gathering Pipelines & Associated Infrastructure	**Doc No.** EA-BFPCL-DSV-GGPL-FEED-GEN-PHY-007R01

The administration of this document is limited to defining the philosophy and DSV's commitment to the Decommissioning and Abandonment (D&A) of the 85 km OKNU-BBOU Gas Gathering Pipeline System and associated infrastructure for the BFPCL Gas Hub / Methanol Plant Project at the end of the asset's operational life.

The document establishes the overarching principles, objectives, and general requirements governing D&A activities, in alignment with prevailing regulatory obligations and recognized industry standards, including those of the Nigerian Upstream Petroleum Regulatory Commission (NUPRC), the Nigerian Midstream and Downstream Petroleum Regulatory Authority (NMDPRA), the Federal Ministry of Environment (FMEnv), and the Environmental Guidelines and Standards for the Petroleum Industry in Nigeria (EGASPIN).

This D&A scope is complemented by the separate 40 km BBOU-ELEP-AKAB down-stream pipeline segment, which operates under an independent Gas Sale and Purchase Agreement (GSPA) and project status, but which interfaces directly with the OKNU-BBOU pipeline system by supplying gas flow for onward transportation to the BFPCL Gas Hub / Gas Processing Plant (GPP). Consequently, the BBOU-ELEP-AKAB segment is expected to undergo decommissioning and abandonment in a broadly similar manner at the end of its service life.

This report has been prepared using data, assumptions, and reference documents provided by the Client. Should additional information become available during subsequent project phases, Epic Atlantic reserves the right to review, amend, or update the opinions, recommendations, and conclusions contained herein.

For completeness and proper interpretation, this document should be read in conjunction with the referenced FEED documents and associated updates listed below, which collectively form integral components of the overall project basis and D&A philosophy.

REFERENCES	
Document No.	**Description**
TBA	Basis of Design Update
TBA	Design and Operating Philosophy Update
TBA	Upstream Gas Gathering Concept Select & Revalidation Report Update
FISAS Draft Report, Nov 2020	ESIA Assessment BBOU-ELEP-AKAB 40 km Gas Supply Project
EA-BFPCL-DSV 0KNU-BBOU-ELEP-AKAB GGPL-FEED-CSM-001	FEED Scope Definition of OKNU-BBOU-ELEP-AKAB Pipeline
Sckope Draft Report, TBA,	Topographical, Geotechnical, and EIA Surveys of OKNU-BBOU, and Updates of BBOU-ELEP-AKAB Pipeline Route Survey
TBA	Relevant As- Built Drawings

Table of Contents

Project Information, Terms, Abbreviations, and Definitions

Project Information – General

Project Information – General		
1.0	Employer	BRASS FERTILIZER & PETROCHEMICAL COMPANY LIMITED (BFPCL)
2.0	PMC	TATA CONSULTING ENGINEERS LIMITED (TCE)
3.0	EPCC (CONTRACTOR)	ENPPI/PETROJET/ENERFLEX CONSORTIUM
4.0	Project Title	Gas Processing and Methanol Complex
5.0	Location	Akabel, Brass Island. The project site is about 85 km South West of Port Harcourt, 70 km South of Yenagoa and 90 km West of Bonny Island.
5.1	State	Bayelsa State, Nigeria
5.2	Nearest Air Port and distance	Port Harcourt Airport (PHC), 80 km Yenagoa Airport (BIA), 75 km
5.3	Nearest Port	Port Harcourt (Onne)
5.4	Nearest Railway Station & distance	-
5.5	Highway and distance	Connected through water way only. Road connection does not exist.
5.6	Important Town and distance	Yenagoa, Nigeria& 67 km
5.7	Altitude Above Mean Sea Level	9 m
5.8	**Atmospheric pressure**	
	Average	**1,013 mbar**
6.0	Ambient Air Temperatures	**(Refer Note-5,6)**
6.1	Maximum	38°C
6.2	Minimum	18°C
6.3	Average	23°C
6.4	Summer Design Wet Bulb Temp	28°C
6.5	Winter Design Wet Bulb Temp	15°C
6.6	MDMT	15°C
6.7	Swamp temperature	25°C (23°C to be used wherever a margin is required)
6.8	Ground temperature	25 - 27.5 °C
6.9	Black-bulb temperature:	80°C
7.0	Relative Humidity	
7.1	Maximum	95 %
7.2	Minimum	51 %
7.3	Average	73 %
8.0	Rainfall Data	

Project Information – General		
8.1	Average annual rainfall	3,800 mm
8.2	Max daily rainfall:	180 mm
8.3	Mean max. Hourly rainfall	100 mm
9.0	Transport	
9.1	Name of the highway near which the plant is located	Connected through water way only. Road connection does not exist.
9.2	Access road	-
9.3	Railway (Gauge)	-
10.0	Wind Data	
10.1	Design Wind Velocity	35.6 m/sec (Note-1)
10.2	Wind Direction	Note-2
11.0	Seismic Data	
11.1	Earthquake Zone	Note-3
12.0	High Flood Level (HFL)	*
13.0	Natural Ground Level	*
14.0	Slope	*
15.0	Drainage	*
16.0	Design temperature for Electrical equipment	40°C

Definition of Terms

TERMS	DEFINITIONS
BFPCL	Brass Fertilizer & Petrochemical Limited.
RENAISSANCE	Renaissance Africa Energy Company Limited
Employer	Party (Usually BFPCL) that initiates the project and ultimately pays for its design and construction. The Employer will generally specify the technical requirements. The Employer may also include an agent or consultant authorized to act for and on behalf of, the Employer.
Engineer	The party that carries out the FEED Engineering of the project.
Manufacturer/Supplier	Party, which manufactures or supplies equipment and services to perform the duties specified by the Contractor.
Shall	The word 'shall' indicate a mandatory requirement.
Should	The word 'should' indicate a recommendation.
Measuring Range	Range defined by LRL and URL within which the limits of uncertainty of the measuring instrument are specified (the capability of the instrument).
On-Line Instruments	Instruments connected to process and utility lines or equipment via primary isolation valves.
Project	Front End Engineering Design for Gas Gathering Pipelines (Phase1)

Abbreviations

Abbreviation	Meaning
AASHTO	American Association of state Highway Transportation Officials
AGEF	Above Ground End Facilities
ALARP	As Low as Reasonably Practicable
API	American Petroleum Institute
ASTM	American Society of Testing and Materials
BOD	Basis Of Design
CP	Cathodic Protection
CPT	Cone Penetrometer Test
CTMS	Custody Transfer Metering Station
DB&B	Double Block and Bleed
DCQ	Daily Contractual Quantity
DEP	Design and Engineering Practice
ESD	Emergency Shutdown
ESDV	Emergency Shutdown Valve
FEED	Front End Engineering Design
FGS	Fire, Gas and Smoke detection systems
GPP	Gas Processing Plant
GTG	Gas Turbine Generators
HAZID	Hazard Identification
HAZOP	Hazard Operability
HEMP	Hazards and Effects Management Process
HFE	Human Factors Engineering
HPU	Hydraulic Power Unit
HRA	Health Risk Assessment
HSES	Health, Safety, Environment and Security
HSSE & SP	Health, Safety, Security, Environment and Social Performance
IEC	International Electro-technical Commission

Abbreviation	Meaning
ISPM	Industry Standard Process Modules
JV	Joint-Venture
LCD	Liquid Crystal Display
LOI	Local Operator Interface
LRL	Lower Range Limit
LRV	Lower Range Value
LIRA	Logistics and Infrastructure Resource Assessment
MAC	Main Automation Contractor
NAG	Non-Associated Gas
OPC	OLE for Process Control
OPPS	Overpressure Protection System
PCD	Process Control Domain
PDMS	Plant Design and Management System
PEPF	Package Engineering, Procurement, and Fabrication
PIG	Pipeline Inspection Gauge
PVC	Polyvinyl Chloride
RTR	Reliability Test Run
SCE	Safety critical elements
SI	International System Units
SIL	Safety Integrity Level
SRB	Sulphate Reducing Bacteria
SIS	Safety Instrumented System
SWB	Steel Wire Braiding
URV	Upper Range Value
USC	US Customary Units
XLPE	Cross Linked Polyethylene
TEG	Tri-ethylene Glycol
TQ	Top Quartile

1. Executive Summary

The scope of this document is limited to defining the philosophy and BFPCL/DSV commitment to the Decommissioning and Abandonment (D&A) of the 85 km OKNU-BBOU Gas Gathering Pipeline System and associated infrastructure for the BFPCL Gas Hub / Methanol Plant Project at the end of the asset's operational life.

This philosophy defines the strategy and execution plan for the safe decommissioning and abandonment of the 85 km OKNU–BBOU gas pipeline network, traversing the swamp and riverine terrain of the Niger Delta.

This D&A scope is complemented by the separate 40 km BBOU-ELEP-AKAB down-stream pipeline segment, which operates under an independent Gas Sale and Purchase Agreement (GSPA) and project status, but which interfaces directly with the OKNU-BBOU pipeline system by supplying gas flow for onward transportation to the BFPCL Gas Hub / Gas Processing Plant (GPP). Consequently, the BBOU-ELEP-AKAB segment is expected to undergo decommissioning and abandonment in a broadly similar manner at the end of its service life, in spite of their being operated in differently operated under different GSPA. In addition, there remains a long-term aspiration to transport up to 1 BSCFD through the integrated pipeline network to a proposed 1 BSCFD gas hub at Akabel under a separate future development, the final configuration and implementation philosophy of which are yet to be defined.

The approach adopts a hybrid strategy (Removal + Abandonment-in-Place) to:

- Ensure zero residual hydrocarbon risk
- Minimize environmental disturbance
- Optimize cost and execution feasibility
- Enable asset recovery (TAR)

The proposal is aligned with project FEED philosophy and regulatory expectations.

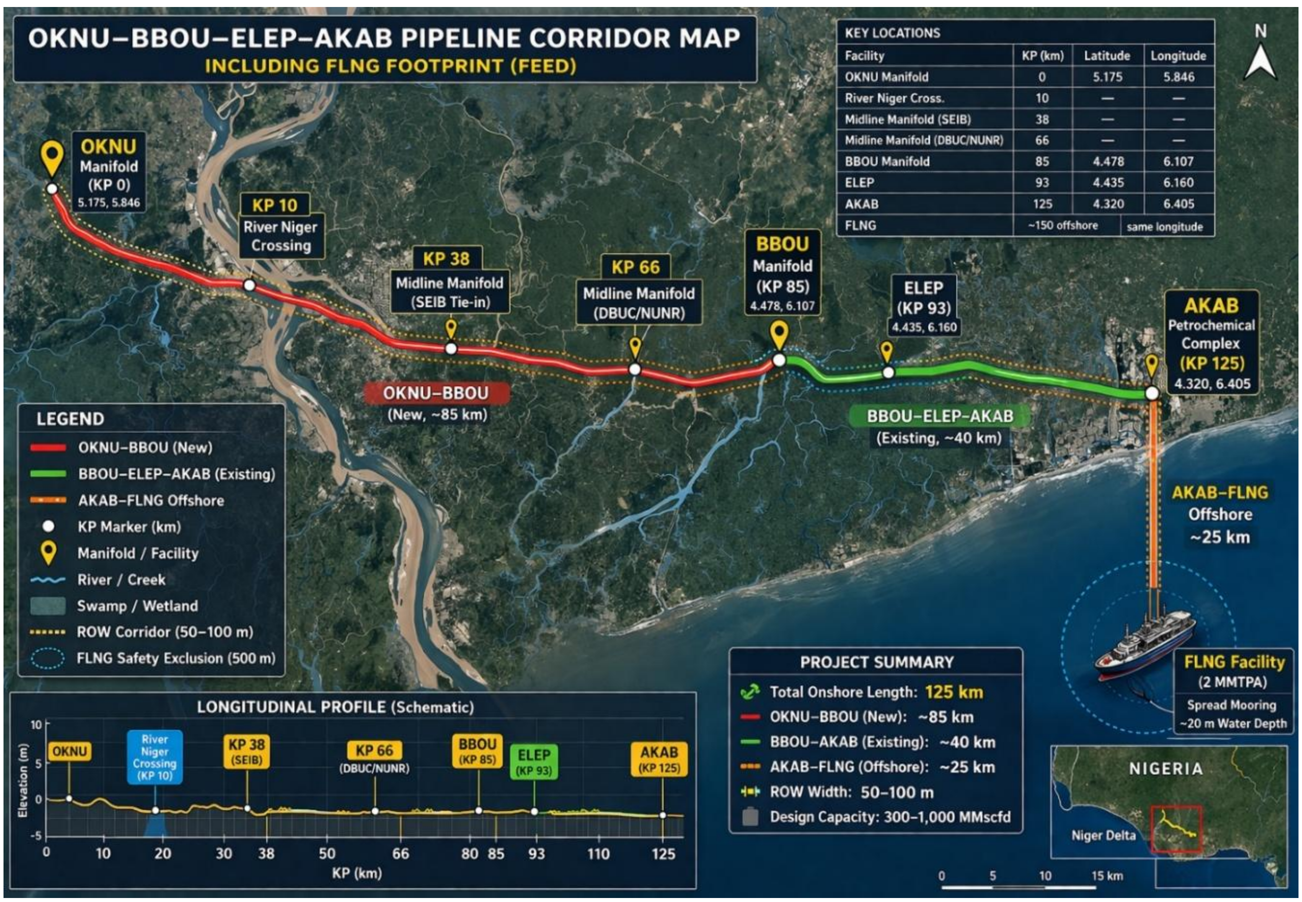
The document establishes the overarching principles, objectives, and general requirements governing D&A activities, in alignment with prevailing regulatory obligations and recognized industry standards, including those of the Nigerian Upstream Petroleum Regulatory Commission (NUPRC), the Nigerian Midstream and Downstream Petroleum Regulatory Authority (NMDPRA), the Federal Ministry of Environment (FMEnv), and the Environmental Guidelines and Standards for the Petroleum Industry in Nigeria (EGASPIN).

While this philosophy document provides a definitive framework for the anticipated D&A approach, the eventual execution strategy at the time of decommissioning shall remain adaptive to evolving regulatory requirements, technological developments, environmental considerations, and emerging industry best practices.

This report has been prepared using data, assumptions, and reference documents provided by the Client. Should additional information become available during subsequent project phases, Epic Atlantic reserves the right to review, amend, or update the opinions, recommendations, and conclusions contained herein.

For completeness and proper interpretation, this document should be read in conjunction with the referenced FEED documents and associated updates listed below, which collectively form integral components of the overall project basis and D&A philosophy.

1.1. System Overview



1.2. BFPCL / DSV Commitment

This proposal sets out a regulator-aligned strategy and reaffirms BFPCL's commitment to the safe decommissioning and abandonment of the 125 km OKNU–BBOU–ELEP–AKAB gas pipeline network in the Niger Delta, of which 85 km OKNU-BBOU Pipeline is part of.

The approach adopts a hybrid methodology—combining Removal and Abandonment-in-Place—to achieve an optimal balance of:

- Safety risk reduction
- Environmental protection
- Cost efficiency
- Asset recovery value

1.2.1. Value Drivers

- Eliminate residual hydrocarbon risk
- Minimize environmental disturbance (swamp/riverine sensitive zones)
- Optimize lifecycle cost and execution efficiency
- Preserve recoverable asset value

1.2.2. Strategic Intent

Hybrid decommissioning approach (Removal + Abandonment-in-Place) delivering safe, compliant, and cost-effective pipeline retirement aligned with FEED philosophy and Niger Delta operating realities.

1.2.3. Governance & Regulatory Framework

- Oversight by NUPRC (upstream) and NMDPRA (midstream/downstream) under the PIA
- Compliance with 2023/2024 Decommissioning & Abandonment Regulations
- Strong host community engagement and satisfaction requirements
- Mandatory Decommissioning & Abandonment Fund (escrow-backed financial assurance) as outlined below:

s/n	Feature	Upstream (NUPRC)	Midstream NMDPRA
01	Primary Law	PIA 2021, Sec 232-233	PIA 2021, Sec 111 & 232
02	Fund Requirement	Bank Escrow (Annual)	Bank Escrow (Annual)
03	Environmental Standard	EGASPIN	NMDPRA Safety Regulations 2023
04	Offshore Rule	IMO Standards (Full removal)	IMO Standards (Full removal)
05	Reporting	36 months before cessation	Varies by license type

1.2.4. Execution Framework

- Early regulatory notification (3–36, depending on scope); and for this scope 12-36 months
  - **Advance Notification:** 12–36 months (system-wide) | ≥3 months (segment-level)
  - **Planning Basis:** 24 months to accommodate regulatory, environmental, and stakeholder approvals
- Comparative assessment
  - **Total Removal vs. Abandonment-in-Place**
- Pipeline hydrocarbon cleaning & purging to EGASPIN standards
- Pipeline sealing, isolation, and stabilization
- Post-abandonment monitoring and RoW integrity management

1.2.5. Key Risks & Mitigations

- Host community acceptance → proactive engagement & Host Community Development Trust (HCDT) alignment
- Vandalism / illegal tapping → grouting, secure isolation, surveillance
- Divestment exposure → strict D&A fund adequacy and operator capability checks

1.3. The Decommissioning & Abandonment Process

The decommissioning and abandonment of the 85 km OKNU–BBOU gas pipeline, at the end of its 30-year service life, will be executed through a **safe, risk-based, and environmentally responsible hybrid strategy (Selective Removal and Abandonment-in-Place)**, tailored to terrain sensitivity and pipeline condition across swamp, riverine, and community areas, comprising:

- System isolation and depressurization, followed by hydrocarbon removal, cleaning, and inerting
- Integrity and environmental assessments to define segment-specific abandonment approaches
- Selective dismantling and recovery of above-ground facilities, crossings, and high-risk sections
- Abandonment-in-place of stable buried segments, with appropriate stabilization and route marking
- Compliance with Nigerian regulatory requirements (NUPRC/NMDPRA, FMEnv, EGASPIN) and international best practices
- Strong HSSE management, stakeholder and host community engagement, and responsible waste handling with emphasis on reuse/recycling

1.3.1. Pipeline Network

- OKNU–BBOU: ~85 km (new design)

Key Nodes

- OKNU Manifold (KP0)
- River Niger Crossing (~KP10)
- Midline Manifolds (KP38, KP66)
- BBOU (KP85)

Terrain

- Predominantly swamp and mangrove
- Multiple river and creek crossings
- Offshore export interface

1.4. Scope Of Decommissioning

The scope covers pipelines, facilities, instrumentation, and ancillary systems across the full 85 km corridor.

1.4.1. Pipelines

- OKNU–BBOU pipeline (~85 km)
- Associated tie-ins and interconnections

1.4.2. Facilities & Process Systems

- Inlet manifold (OKNU)
- Midline manifolds (KP38, KP66)
- BBOU manifold
- AKAB gas processing hub interface
- Slug catcher, if any
- Pigging facilities (launchers/receivers)

1.4.3. Metering Systems (Custody Transfer & Verification)

The pipeline network incorporates **metering facilities at all manifolds and the gas plant outlet**, forming a critical component of the decommissioning scope.

Wet Gas Meters (All Manifolds)

Installed at:

- OKNU (KP0)
- KP38
- KP66
- BBOU (KP85)

Function:

- Measurement of wet gas streams (with entrained liquids)
- Flow monitoring and allocation

Decommissioning Scope:

- Isolation and depressurization
- Removal of metering skids and instrumentation
- Preservation for reuse (subject to integrity verification)

1.4.4. Check Meters (All Manifolds)

Installed at:

- OKNU (KP0)
- KP38
- KP66
- BBOU (KP85)

Function:

- Independent flow verification
- System balancing and reconciliation

Decommissioning Scope:

- Controlled shutdown under monitored conditions
- Final calibration validation and reconciliation
- Removal of meter runs and associated instrumentation
- Data archiving for audit purposes

The strategy adopts a **hybrid approach (Removal + Abandonment-in-Place)** to:

- Eliminate residual hydrocarbon risk
- Minimize environmental disturbance
- Optimize lifecycle cost

- Preserve recoverable asset value

The approach is aligned with FEED philosophy, regulatory expectations, and Niger Delta execution realities.

1.5. Indicative D&A Schedule

Phase	Activity	Indicative Duration
1	Regulatory notification and stakeholder engagement	3 months
2	Detailed engineering and D&A planning	4 months
3	Environmental and statutory approvals	3 months
4	Pipeline isolation, depressurization, and cleaning	2 months
5	Removal of above-ground facilities	3 months
6	Pipeline abandonment/removal activities	5 months
7	Waste management and disposal	2 months
8	Environmental remediation and restoration	4 months
9	Final inspection, certification, and close-out	2 months

Total Indicative Project Duration

Approximately 18–24 months, depending on logistics, seasonal access constraints, regulatory processing timelines, and environmental conditions.

1.6. Preliminary USD Cost Estimate

Item	Description	Estimated Cost (USD Million)
1	Engineering studies and D&A planning	1.20
2	Regulatory compliance and permitting	0.60
3	Pipeline cleaning, gauging, and flushing	2.50
4	Isolation and depressurization works	0.80

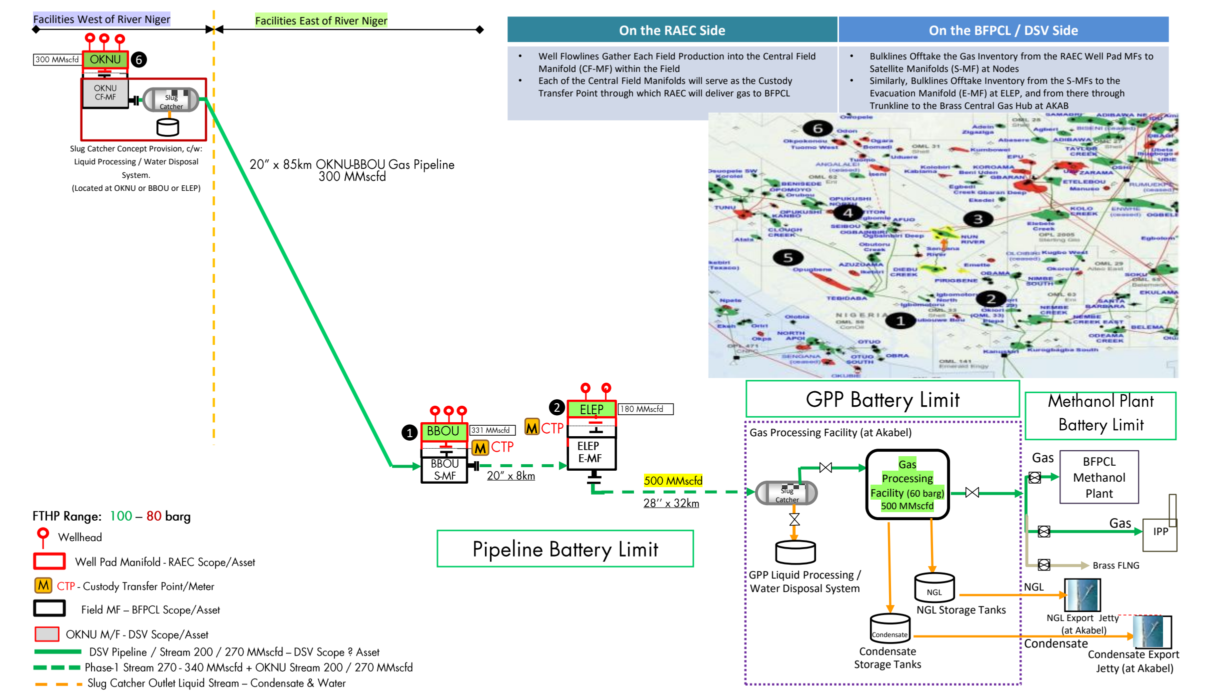
5	Removal of above-ground facilities	3.20
6	Selective pipeline removal works	5.50
7	Abandonment-in-place activities	2.80
8	Waste handling, transportation, and disposal	1.60
9	Environmental remediation and restoration	3.80
10	Marine/swamp logistics and access support	2.20
11	Project management and supervision	1.50
12	HSE management and security support	1.10
13	Contingency allowance (15%)	3.70
Total Preliminary D&A Cost Estimate		**USD 28.50 Million**

2. Introduction & Background

2.1. Base Case Gas Gathering System

Network of pipelines and manifolds transporting gas from **OKNU, BBOU, and ELEP fields** to the central processing facility.

Schematics of Gas Supply from Okpokunou alongside Phase-1 Gas Supply



Satellite Imagery of Gas Supply from Okpokunou alongside Phase-1 Gas Supply

2.2. Gas Processing Plant (GPP) – 500 MMscfd

- Extraction of Natural Gas Liquids (NGLs)
- Delivery of lean gas to downstream facilities, Includes:
  - Common front-end processing
  - Dedicated polishing units for Methanol and LNG

2.3. Methanol Production Plant

- Two trains, each rated at 5,000 MTPD
- Total production capacity: 10,000 MTPD

2.4. Floating LNG (FLNG) Facility/Fertilizer (Urea)

- Capacity: ~2 MMTPA LNG
- Capacity: ~2.8MMTPA UREA
- Feed gas: >200 MMscfd processed gas
- Location: ~20 km offshore, ~20 m water depth, complete with +25 km Feed Gas Supply Line from Gas Hub at AKAB to the FLNG/Fertilizer
- Products:
  - LNG/Urea
  - Condensate
  - LPG (Butane & Propane)

2.5. Product Export System

- Export pipelines
- Offshore Single Buoy Mooring (SBM) system for methanol and NGL export
- Dry port

2.6. Produced Water Management

- Handling, treatment, and disposal facilities
- To be defined through **Decision Concept Selection (DCS)** studies

2.7. Utilities and Offsite Infrastructure

- Power generation systems
- Steam generation
- Product storage facilities
- Export jetty and marine infrastructure
- All required off-plot and support systems

2.8. Pipeline Design Philosophy – Flexibility & Expandability

While the primary objective of the OKNU–BBOU pipeline is evacuation of Okpokunou gas production, the design incorporates provisions for **future network expansion** through strategically located **midline manifolds**, which will also function as **sectionalizing valve stations**.

Planned Tie-in Points:

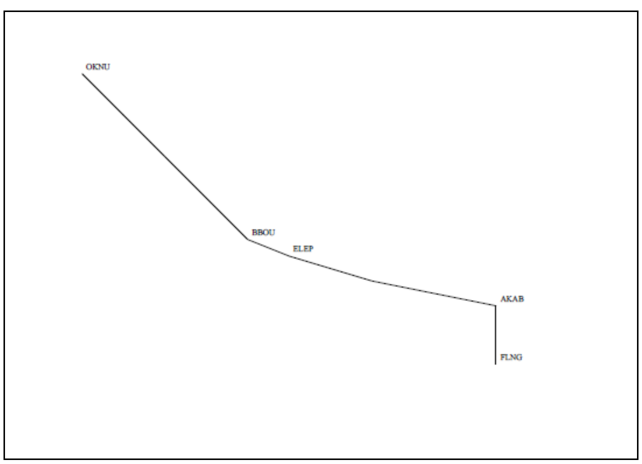
- **KP 38, - for tie-in of SEIB production:**
    - Confirmed tie-in point
    - Identified as a key gas supply node
- **KP 66, - for tie-in of NUNR / DBUG production:**
    - Conditional tie-in, subject to future development

Given proximity and routing efficiency, **SEIB** is considered the most practical near-term tie-in to the OKNU–BBOU pipeline, hence the midline manifold at KP 38

For **NUNR and DBUC**, a shared midline manifold near DBUC, subject to firm gas supply commitments, will be more cost effective than dedicated manifolds. The option of an **independent pipeline network to gather NUNR & DBUC production to BBOU** or ELEP will be evaluated during the FEED.

The Facilities Coordinates

Location	Latitude	Longitude	KP (km)
OKNU Manifold	5.175	5.846	0
River Niger Crossing at KP 10			10
Midline Manifold at KP 38, near SEIB			38
Midline Manifold at KP 66, near DBUC			66
BBOU Manifold	4.478	6.107	85
ELEP	4.435	6.16	93
AKAB	4.32	6.405	125
FLNG/Fertilizer	25 km due south of AKAB (same longitude)		



3. Battery Limit, and Scope of Work

3.1. Battery Limit

The phase 1 facilities are the focus of the present FEED, however with consideration for provisions and retrofits that will accommodate the phase 2 facilities in a rather seamless manner.

The Phase-1 main facilities are:

- OKNU – BBOU pipeline, 85 km

The battery limits are shown below:

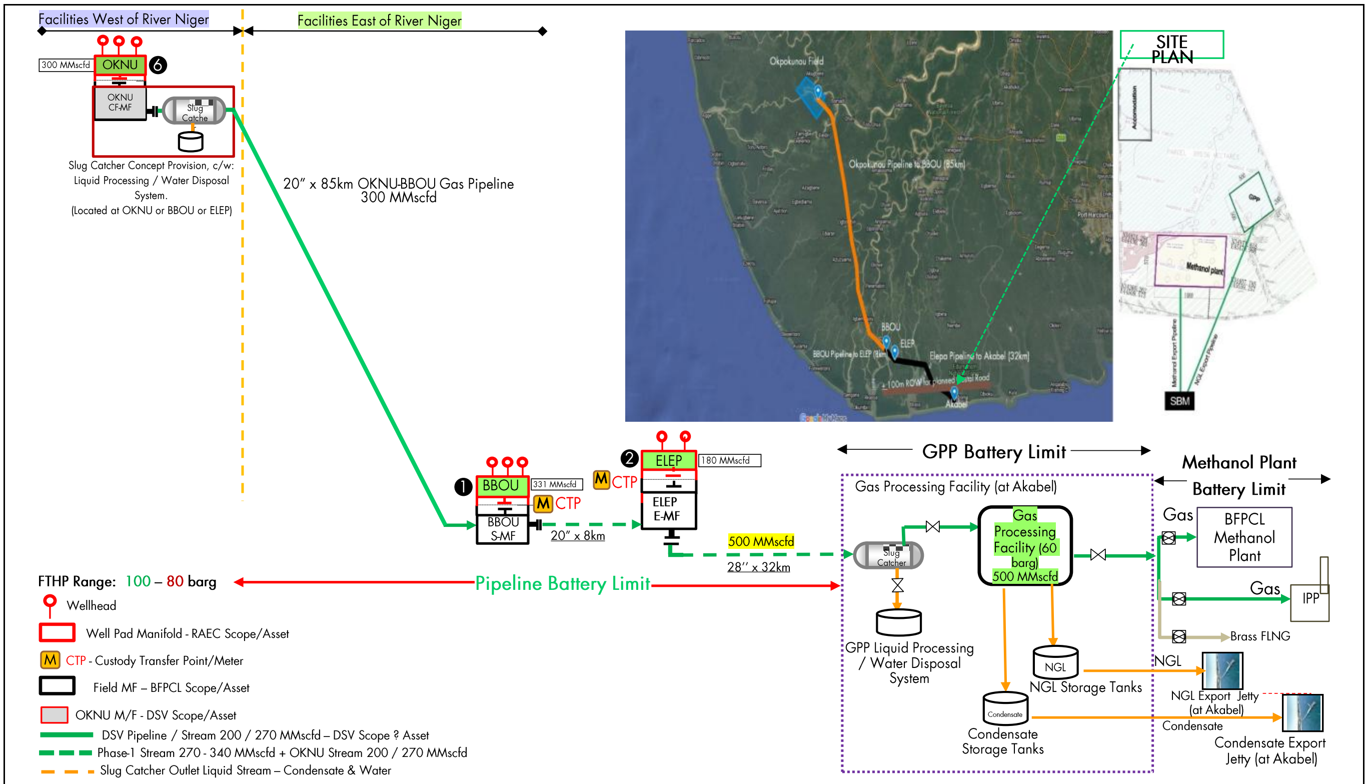


Figure 3.1 Battery Limit – Pipeline

3.1.1. OKNU Platform

- The terminal flanged ball valve on the manifold collecting well fluids from OKNU Wells.
- The flanged ball valve provided for the tie-in of OKNU Pipeline on the manifold transporting well fluids to BBOU.

3.1.2. BBOU Platform

- The terminal flanged ball valve on the manifold collecting well fluids from BBOU Wells.
- The flanged ball valve provided for the tie-in of Opugbene (OPUG) Pipeline on the manifold transporting well fluids to ELEP.

3.2. Data Room Index Structure

3.2.1. Project Overview

- Pipeline description
- Key coordinates and KP definitions
- Asset register

3.2.2. Route & Survey

- Alignment sheets
- ROW drawings
- Survey as-builts

3.2.3. Pipeline Engineering

- GA drawings
- Isometrics
- Wall thickness & specs

3.2.4. Facilities

- OKNU
- KP38 Midline manifold
- KP66 Midline manifold
- BBOU

3.2.5. Crossings

- River Niger crossing
- Creek crossings
- HDD drawings

3.2.6. Mechanical & Materials

- Weld maps
- Material certificates
- Coating specs

3.2.7. Process & Instrumentation

- P&IDs
- Cause & Effect
- SCADA

3.2.8. Electrical & CP

- CP layouts
- SLDs
- Earthing

3.2.9. Civil & Structural

- Foundations
- Supports
- Access infrastructure

3.2.10. Testing & Commissioning

- Hydrotest
- Drying
- Pre-commissioning

3.2.11. Environmental

- EIA
- Baseline data
- Restoration plans

3.2.12. Modifications

- Redlines
- Tie-ins
- Design changes

3.2.13. Decommissioning Documents to be developed)

- Decommissioning Execution Plan
- Cleaning & pigging procedures
- Waste management plan
- Restoration plan

3.3. Layout & Configuration

3.3.1. Route & Alignment

- Pipeline Alignment Sheets (Plan & Profile, KP 0–85 km)
- ROW Drawings (full corridor coverage)
- Survey As-Built (final coordinates and deviations)
- Pipeline General Arrangement (GA)
- Plot Plans:

    - OKNU Manifold (KP 0)
    - BBOU Manifold (KP 85)
- Pipeline Isometrics (segment-wise)

3.3.2. Crossings & Special Sections

- River Niger Crossing (~KP 10) drawings
- Creek/River crossings (all locations)
- HDD / trenchless installation drawings
- Trench and burial profiles

3.3.3. Mechanical & Materials

- Weld maps (full traceability)
- Wall thickness and coating specs
- Line pipe data sheets
- Material certificates (API 5L traceability)

3.3.4. Valves & Facilities

- Valve location drawings (chainage-based)
- Pig launcher/receiver drawings
- Manifold piping layouts (OKNU, KP38, KP66, BBOU, ELEP)
- Slug catcher drawings

3.3.5. Process & Control

- As-built P&IDs (entire system)
- Cause & Effect diagrams
- SCADA / instrumentation layouts

3.3.6. Electrical & CP

- Cathodic Protection layout
- Transformer Rectifier (TR) locations
- Earthing & bonding drawings
- Electrical Single Line Diagrams

3.3.7. Civil & Structural

- Foundations (manifolds, slug catcher if any)
- Pipe supports and sleepers
- Access roads and pads

3.3.8. Testing & Commissioning

- Hydrotest layouts and sectioning
- Dewatering/drying layouts

3.3.9. Environmental

- Baseline environmental maps
- Sensitive area mapping (mangroves, wetlands)
- Restoration drawings (if applicable)

3.3.10. Modifications & Deviations

- Redline drawings
- Tie-in modifications (KP38, KP66, BBOU integration)
- Looping / resizing updates

3.4. Submission Requirements

- Digital format (PDF + native CAD)
- Indexed and searchable
- Linked to asset register
- Verified as "As-Built" (not IFC/IFD)

3.5. KP-by-KP Drawing Relevance (OKNU → BBOU)

This maps **critical drawings to physical pipeline segments**—very useful for execution planning.

3.5.1. KP 0 – KP 10 (OKNU → River Niger Crossing)

Key Features: Inlet manifold, slug catcher, swamp terrain

- Required Drawings:
- OKNU Plot Plan & P&IDs
- Slug catcher, if any, GA
- Alignment sheets
- Valve and pigging drawings
- CP layout

3.5.2. KP 10 – River Niger Crossing

Key Features: Major river crossing (~200 m width)

- Required Drawings:
- HDD / crossing profile drawings
- Bathymetric and geotechnical drawings

- Crossing method as-built
- Environmental sensitivity maps

Execution Note: High likelihood of **abandonment-in-place**

3.5.3. KP 10 – KP 38 (Swamp Section)

Key Features: Challenging swamp terrain

- Required Drawings:
- Alignment + trench details
- CP and coating drawings
- Weld maps
- Access road drawings

3.5.4. KP 38 – Midline Manifold (provisional for SEIB Tie-in)

Key Features: Future tie-in node

- Required Drawings:
- Manifold GA and piping layout
- Metering system drawings
- Isolation valve drawings

3.5.5. KP 38 – KP 66

Key Features: Long swamp stretch

- Required Drawings:
- Alignment sheets
- Weld maps
- CP layout
- Environmental maps

3.5.6. KP 66 – Midline Manifold (NUNR/DBUC Tie-in)

Key Features: Conditional tie-in

- Required Drawings:
- Manifold GA
- Tie-in spool drawings
- Instrumentation layouts

3.5.7. KP 66 – KP 85 (BBOU)

Key Features: Approach to major hub

- Required Drawings:
- Alignment and profile
- Valve segmentation drawings
- Hydrotest section drawings

3.5.8. KP 85 – BBOU Manifold

Key Features: Major integration node

- Required Drawings:
- Plot plan & P&IDs
- Tie-in drawings (OKNU + ELEP line)
- Metering and control systems

4. The Decommissioning & Abandonment Scope

4.1. Governance

The exercise / process is governed be by commitment and host community satisfaction through stakeholder engagement

4.1.1. Primary Regulators

- NUPRC (Nigerian Upstream Petroleum Regulatory Commission): Oversees pipelines connected to production fields and offshore platforms.
- NMDPRA (Nigerian Midstream and Downstream Petroleum Regulatory Authority): Oversees trunk lines, distribution networks, and processing plant connections.

4.1.2. Core Legal Frameworks

Both regulators have issued specific 2023/2024 regulations derived from the PIA:

- Upstream: Nigerian Upstream Petroleum Decommissioning and Abandonment Regulations, 2023 (and 2024 amendments).
- Midstream/Downstream: Midstream and Downstream Decommissioning and Abandonment Regulations, 2023.

The Decommissioning & Abandonment Fund (The "Escrow")

Under the PIA, every licensee must set up a D&A Fund.

- Mandatory Contribution: You must contribute annually to an escrow account held in a commercial bank.
- Purpose: This ensures that even if a company goes bankrupt, the Nigerian government has the funds to safely remove the infrastructure.
- 2024 Update: Recent amendments allow for a portion of these funds (up to 15% for IOCs) to be held in local banks, while the rest can be in rated international institutions.

4.1.3. Technical & Operational Requirements

The NUPRC and NMDPRA follow a "Comparative Assessment" model similar to the UK, but with a heavy focus on the Environmental Guidelines and Standards for the Petroleum Industry in Nigeria (EGASPIN).

Standard Procedure for Pipelines:

a. Notice of Intent: You must notify the regulator at least 3 months (for specific line segments) to 36 months (for entire fields/systems) before the planned start.
b. Comparative Assessment: You must provide a study comparing:

- Total Removal: Digging up and removing the pipe.
- Abandonment in Place: Cleaning, capping, and leaving the pipe buried.

c. Cleaning & Purging: Pipelines must be flushed of all hydrocarbons using water or inert materials until they meet EGASPIN's "zero-sheen" or ppm standards.

d. Sealing: Pipes must be capped at both ends with moisture-resistant materials.

e. Post-Abandonment Monitoring: For pipelines left in place, the operator often remains liable for the Right of Way (RoW) and must ensure no "subsidence" (ground sinking) occurs over time.

4.1.4. Key Challenges in the Nigerian Context

- Host Community Involvement: The PIA requires operators to consult with Host Community Development Trusts. You must prove that abandoning a pipeline in place won't interfere with local farming, fishing, or land use.
- Security & Vandalism: Abandoned pipelines that aren't properly "grouted" (filled with concrete) are often targeted by illegal refiners who believe residual oil remains, leading to significant environmental spills long after the line is "closed."
- Divestment Issues: If a major international oil company (IOC) sells a pipeline to a local firm, the NUPRC now strictly audits the D&A Fund to ensure the new owner can afford the eventual cleanup.

4.1.5. Ancillary Systems

- Cathodic protection systems
- Electrical and SCADA systems
- Civil foundations and supports
- River and creek crossing structures

4.2. Decommissioning Philosophy

- Safety-first execution (zero harm, zero release)
- Environmental protection (no inland discharge)
- Regulatory compliance
- Value optimization (TAR)
- Minimum disturbance in swamp terrain

4.3. Key Decommissioning Processes

4.3.1. Planning & Approvals

- Asset verification
- Environmental baseline
- Decommissioning Execution Plan (DEP)
- Regulatory and stakeholder engagement

4.3.2. Shutdown & Isolation

- Controlled shutdown
- Mechanical and electrical isolation
- Segmentation into manageable sections

4.3.3. Cleaning & Hydrocarbon Removal

- Depressurization
- Pigging and residue removal
- Sludge and condensate handling

4.3.4. Flushing, Drying & Inerting

- Flushing (water/chemical)
- Drying (air/nitrogen)
- Nitrogen inerting

4.3.5. Metering Close-Out (Critical Step)

- Final measurement reconciliation
- Calibration validation
- Data archiving and audit closure

4.3.6. Disconnection & Segmentation

- Pipeline cutting and isolation
- Disconnection at all manifolds and facilities

4.3.7. Execution (Removal / Abandonment)

- Segment-specific execution based on terrain and risk

4.3.8. Waste Management

- Segregation and certified disposal

4.3.9. Site Restoration

- Backfilling and grading
- Mangrove restoration

- Environmental monitoring

4.4. Abandonment Activities

- Cleaning to hydrocarbon-free condition
- Permanent isolation (caps/blinds)
- Filling (water / grout / seawater)
- CP system decommissioning
- Long-term integrity assurance
- GIS updates and documentation

4.5. KP-Based Execution Strategy

S/N	Segment	Strategy	Justification
1	KP0 (OKNU)	Remove	High-value facilities (TAR)
2	KP0–10	Mixed	Accessible sections removable
3	KP10 (River Niger)	Abandon	High environmental risk
4	KP10–38	Abandon (majority)	Deep swamp
5	KP38 Manifold	Remove	Strategic node
6	KP38–66	Mixed	Access-dependent
7	KP66 Manifold	Remove	Future tie-in
8	KP66–85	Mixed	Near infrastructure

4.6. Risk Management

Risk	Mitigation
Residual hydrocarbons	Multi-stage cleaning + inerting
Environmental impact	Abandonment strategy
Swamp instability	Specialized equipment
Regulatory delays	Early engagement
Commercial disputes	Metering close-out

4.7. Conclusion

This proposal delivers a technically robust, environmentally responsible, and commercially optimized framework for decommissioning the OKNU–BBOU pipeline system.

It ensures:

- Safe and complete hydrocarbon removal
- Minimal environmental impact
- Full regulatory compliance
- Optimized cost through hybrid execution
- Preservation of asset value

5. Key Decommissioning Processes & Abandonment Activities

5.1. Decommissioning Processes (End-to-End Execution)

5.1.1. Pre-Decommissioning Planning

- Asset verification (as-built vs actual condition)
- Regulatory approvals (Nigerian Upstream Petroleum Regulatory Commission, Nigerian Mid-stream and Downstream Petroleum Regulatory Authority)
- Environmental baseline and risk assessment
- Stakeholder/community engagement

5.1.2. System Shutdown & Isolation

- Controlled shutdown of gas flow
- Mechanical isolation (valves, blinds, spading)
- Electrical and instrumentation de-energization
- Segmentation into manageable sections (by KP / hydrotest sections)

5.1.3. Pipeline Cleaning & Hydrocarbon Removal

- Depressurization and venting
- Pigging operations:
  - Cleaning pigs
  - Batching pigs
- Chemical cleaning (if required)
- Removal of condensate, sludge, and residues

5.1.4. Flushing, Drying & Inerting

- Water flushing or gel cleaning
- Air or nitrogen drying
- Nitrogen purging to achieve inert condition

5.1.5. Disconnection & Mechanical Works

- Cutting of pipeline into sections
- Disconnection from:
  - Manifolds (OKNU, BBOU)
- Removal of above-ground facilities:
  - Slug catcher, if any

- Pigging stations
- Metering systems

5.1.6. Pipeline Removal (Where Applicable)

- Excavation (swamp-compatible methods)
- Lifting and transport
- Removal of:
  - Shallow buried pipelines
  - Congested corridor sections
- Special handling at crossings

5.1.7. Waste Management

- Segregation:
  - Scrap steel
  - Hazardous waste
  - Contaminated soils
- Transport to certified disposal facilities
- Compliance with Federal Ministry of Environment standards

5.1.8. Site Restoration

- Backfilling and grading
- Mangrove and vegetation restoration
- Erosion control (especially in swamp areas)
- Post-restoration monitoring

5.2. Abandonment Activities (Pipeline Left In Place)

Applied where removal is technically difficult, environmentally risky, or uneconomic.

5.2.1. Cleaning to Acceptable Standard

- Internal cleaning to remove hydrocarbons
- Verification via pigging and sampling

5.2.2. Isolation & Disconnection

- Physical disconnection from live systems
- Installation of:
  - Blind flanges
  - Permanent caps

- Removal of surface tie-ins

5.2.3. Filling / Stabilization

- Depending on location:
- Onshore / Swamp
- Filling with inert material:
    - Nitrogen (short-term)
    - Water or inhibited fluid
    - Cement grout (long-term stability)
- River / Creek
- Flooding with Creek Water
- Natural sediment stabilization
- Rock dumping (if required)

5.2.4. Cathodic Protection Decommissioning

- Deactivation of CP systems
- Removal or safe abandonment of anodes
- Electrical isolation

5.2.5. Long-Term Integrity Assurance

- Corrosion risk assessment
- Monitoring (if required by regulator)
- Documentation of abandoned sections

5.2.6. Marking & Records

- Updating pipeline records as "abandoned"
- GIS and survey updates
- Installation of surface markers (where required)

5.3. Special Considerations for OKNU–BBOU

- High-Priority Abandonment Zones
- River Niger crossing (~KP10) → Abandon in place preferred
- Deep swamp sections (KP10–KP66) → Selective abandonment
- High-Priority Removal Zones
- Manifold areas (OKNU, KP38, KP66, BBOU)
- Congested corridors
- Areas with future development conflicts

6. Facilities Engineering Design Features

6.1. Pipeline Routing Philosophy

The original concept for the Okpokunou Pipeline envisaged a direct connection to the Elepa Manifold (ELEP). However, subsequent routing assessments indicated that the proposed alignment would intersect the BBOU manifold corridor and run in close parallel with the existing/planned BBOU–ELEP pipeline in the downstream section.

It is therefore considered technically and economically prudent to terminate the pipeline at the **BBOU manifold** and utilize the planned BBOU–ELEP pipeline segment rather than constructing a dedicated line to ELEP.

This approach offers the following advantages:

- Approximate reduction of ~ 8 km in pipeline length
- Measurable CAPEX savings
- Reduced environmental footprint
- Minimized corridor congestion
- Improved constructability within Niger Delta swamp terrain

6.2. Liquid Management Strategy

6.2.1. Upstream Liquid Removal

The base case concept provides for removal of the majority of entrained liquids upstream at Okpokunou in order to:

- Reduce hydraulic loading
- Improve flow stability
- Minimize slugging risk
- Optimize downstream pipeline sizing

Accordingly, a **slug catcher** is proposed at Okpokunou.

6.2.2. Diameter Optimization

Early liquid removal may permit optimization of pipeline diameter. However, final sizing will be confirmed during FEED through:

- Steady-state hydraulic modeling
- Transient multiphase simulation
- Comparative evaluation of alternate slug catcher locations (inlet vs midline vs outlet)
- Sensitivity cases for turndown and ramp-up scenarios

6.3. Receiving Facility Protection

A slug catcher at the Pipeline End Manifold (PLEM) will provide protection against liquid slugs, thereby safeguarding downstream processing equipment and ensuring stable plant operations.

6.4. Liquid Handling and Disposal

The slug catcher facility will include full liquid-handling systems comprising:

- Condensate/NGL transfer facilities to AKAB for processing
- Produced water collection and handling system
- Storage and metering as required
- Produced water disposal options will include:
  - Offshore discharge (subject to regulatory compliance)
  - De-mineralisation for industrial use
  - Other approved beneficial reuse options

Discharge into inland waters is not permissible under **NUPRC** and **FMEnv** regulations.

6.5. Pipeline Capacity and System Integration

The pipeline will be designed to transport up to **300 MMscfd** of raw wellhead natural gas, made up as follows:

1. Design Throughput

The pipeline will be designed for a maximum capacity of 300 MMscfd of raw wellhead natural gas, comprising:

- **200 MMscfd** from the **RAEC-DSV** Gas Sales and Purchase Agreement
- ***70 MMscfd make-up gas*** on RAEC 270 MMscfd, in case the 70 MMscfd production will be coming from the fields whose production will be tied-in to the Okpokunou pipeline, or even from Okpokunou itself.
- **30 MMscfd design allowance,** - for production upside
- Total Design Capacity: **300 MMscfd**

The 300 MMscfd will flow to the BBOU manifold, from where the gas will join the BBOU enroute to ELEP via an existing/planned 20” x 8km to the ELEP M/F, enroute through 28-inch × 32 km trunkline to the Akabel Gas Processing Hub. At Akabel, the gas will be processed as feedstock for a Methanol Manufacturing Plant.

6.6. System Integration

Gas will flow:

- From OKNU to BBOU via the proposed 20" x ~ 85 km pipeline

The OKNU–BBOU pipeline shall therefore be designed to reliably transport up to **300 MMscfd** under all operating scenarios, while maintaining full flow assurance across the integrated **OKNU–BBOU–ELEP–AKAB** pipeline network, ensuring delivery of the target **500 MMscfd** to the Gas Hub.

6.7. Pipeline Routing and Terrain

The approximately 85 km pipeline will traverse challenging Niger Delta swamp terrain and include:

- One major crossing of the **River Niger,** (a distributary of the **Niger River**), estimated at:
    - 200 m width
    - 9 m water depth
- Four (4) additional river crossings
- Eight (8) creek crossings
- Standard inlet manifold facilities at OKNU
- Terminal manifold facilities at BBOU

Special crossing methodologies (e.g., HDD, microtunneling, or cofferdam/trench methods) will be evaluated during FEED based on geotechnical and bathymetric investigations.

6.8. Pipeline Size Decision

The pipeline diameter shall be selected as the **optimum commercial size** capable of reliably transporting the forecast gas volumes to BBOU at a delivery pressure sufficient to ensure that the combined streams from BBOU arrive at the Gas Plant inlet at the required **60 barg**.

The final diameter will be established through integrated hydraulic and flow assurance modelling of the full export corridor, including:

- OKNU–BBOU pipeline (new design subject line, currently assessed at 20"
- Existing Design 20" BBOU–ELEP pipeline, currently assessed at 20" – 24"
- Existing Design 28" ELEP–AKAB pipeline, currently assessed at 28" – 30"

System-wide validation will confirm that the selected diameter sustains the target throughput while maintaining pressure, temperature, and operability margins across all operating scenarios. Should modelling indicate hydraulic or flow assurance constraints within the downstream segments, re-sizing of the affected sections will be evaluated to preserve system integrity and performance.

Where the hydraulically optimal diameter differs materially from the commercially efficient size, competing options shall be comparatively assessed based on:

- Capacity robustness and future expansion potential
- Flow assurance risks (liquids handling, turndown, slugging, hydrate margin)
- CAPEX and lifecycle OPEX implications

- Constructability and swamp terrain constraints
- Uncertainty in production forecast and ramp-up profile

The selected diameter will therefore represent a balanced, risk-informed decision that optimizes commercial viability, operational reliability, and long-term system flexibility.

6.9. Metering System

6.9.1. Wet Gas Metering (For Hydrocarbon Accounting)

Hydrocarbon accounting will primarily be based on the contractual RAEC wet gas meters, to which BFPCL is granted full ("unfettered") access under the GSPA, including integration into its Control System.

To ensure measurement assurance and data validation, BFPCL will deploy a limited number of independent check meters. These are discretionary and intended solely for reconciliation purposes, not custody transfer.

The optimized check metering philosophy is as follows:

- **Elepa (Pipeline Inlet):** One check meter installed at pipeline entry (Pipeline FEED scope).
- **Akabel (Pipeline Outlet):** One check meter at pipeline discharge to the slug catcher (Pipeline FEED scope).
- **GPP Inlet (Slug Catcher):** Check meters at both gas and liquid legs feeding the GPP train (GPP FEED scope).

6.9.2. The Fiscal Metering system

The wet gas meters at delivery points form part of the overall fiscal framework. However, fiscal metering in this context specifically refers to **processed (lean) gas and associated fluids**.

- Fiscal metering facilities (gas, condensate, and produced water) are designated for RAEC but will be **located within the GPP** and **jointly operated by BFPCL and RAEC**, in accordance with the GSPA.
- Meter duplication is to be avoided; therefore, any additional BFPCL-installed meters will be strictly for **reconciliation and performance monitoring** (e.g., simple mass flow meters).
- Laboratory data from BFPCL will be integrated into the reconciliation process to enhance accuracy and auditability.

6.9.3. Other Metering Systems

BFPCL will independently design and install metering systems for gas streams allocated to **methanol production and utilities**.

- The preferred metering technology is **ultrasonic flow metering**, capable of accurately determining both **flow rate and energy content** at plant entry.
- Final selection and inclusion within the GPP vendor scope will be confirmed during detailed engineering.

6.9.4. Governance & Integration

- A **Joint Operations Team (Upstream–Midstream interface)** will be established, covering the full chain from wellhead to GPP, in line with the GSPA.
- All metering systems included in the GPP bid will be **reviewed and rationalized** to ensure clarity of purpose and eliminate redundancy.